

A Smart Traffic Light Modeling Using AnyLogic Simulation

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Abstract— Urban traffic congestion remains a persistent issue in modern cities, often exacerbated by traditional traffic signal systems that operate on static cycles, disregarding real-time traffic conditions. This study presents a dynamic and smart traffic light control model using AnyLogic simulation software, leveraging discrete event simulation principles. The proposed model simulates a three-way urban intersection with directional flows from the west, east, and south, incorporating both car and bus agents governed by adaptive signal timing logic. By integrating the Road Traffic Library in AnyLogic, the system evaluates traffic density and adjusts green light durations based on vehicle queue and flow patterns. The simulation environment enables the modeling of complex behaviors such as lane selection, queuing, and pedestrian crossing interactions. Results demonstrate improved intersection efficiency through reduced vehicle wait times and balanced traffic flow across all directions. The 3D visualization further enhances the validation of traffic behavior and signal phase coordination in real-time. The findings emphasize the potential of simulation-based models as decision support tools in designing smart transportation systems geared towards smart cities.

Keywords—AnyLogic, simulation model, traffic lights, traffic flow, vehicle

I. INTRODUCTION

Urban traffic congestion is a growing concern, especially in densely populated cities, where inefficient traffic signal systems contribute significantly to travel delays, environmental pollution, and economic losses [1]. Traditional traffic signal systems typically operate on pre-set cycles, often failing to adapt to real-time traffic fluctuations. Intelligent Transportation Systems (ITS), powered by advanced computing and communication technologies, offer promising solutions to modernize traffic flow control. Within this context, discrete event simulation (DES) modelling [2] has emerged as an effective technique to replicate complex, dynamic traffic

systems and provide deeper insights into vehicular movement and signal coordination. AnyLogic, a powerful hybrid modelling tool, allows the integration of DES, system dynamics, and agent based modelling to build realistic and scalable traffic simulation models. Studies have demonstrated the value of simulation-based strategies [3] in improving intersection performance, reducing vehicle delay, optimizing traffic light phases [4], and accommodating pedestrian flows. By incorporating tools such as the Road Traffic Library in AnyLogic, researchers can create micro-level traffic models that closely represent real-world behaviour. This paper focuses on developing a smart traffic light simulation model using AnyLogic to analyse and optimize signal timings at intersections, thereby improving traffic efficiency and reducing congestion. Such models are essential for future smart city infrastructures aimed at sustainable urban mobility.

Traffic congestion at urban intersections is a critical challenge, especially during peak hours, due to static and non-adaptive traffic light systems [5]. These systems allocate green signal timings based on historical or average traffic data, ignoring real-time variations in traffic density across lanes [6]. As a result, some roads experience excessive delays while others waste valuable green time. This imbalance causes increased travel times, fuel consumption, and CO₂ emissions, leading to environmental degradation and economic inefficiencies. Conventional approaches often fall short in dynamically optimizing signal timing based on traffic load [7]. Moreover, small cities and municipalities may lack the financial and infrastructural capacity to deploy high-end traffic management solutions, making scalable and cost-effective models necessary [8]. Hence, there is a need for an intelligent, adaptable traffic control system that can analyze current traffic conditions and adjust signal timings accordingly. The use of simulation platforms like AnyLogic [9][10] allows researchers to model these systems virtually, test multiple configurations, and derive optimal solutions. This paper aims to address the limitations of

fixed-time traffic signal systems and develop a dynamic model that adapts signal phases to real-time traffic variations using AnyLogic simulation tools.

The objective of this research is to model and simulate an intelligent traffic light system using DES in AnyLogic software that dynamically adjusts signal timings based on real-time traffic conditions. The model aims to reduce congestion, vehicle waiting time, and improve traffic flow efficiency at intersections. The research employs AnyLogic's Road Traffic Library to build a DES traffic light model of an urban intersection [11]. Vehicles are modelled as agents with defined behaviours such as speed, route selection, and lane changes. The model incorporates traffic sensors and signal control logic to dynamically adjust green signal durations based on vehicle queue lengths. Data inputs [12] include vehicle arrival rates, intersection geometry, and signal phase logic. The model undergoes calibration of traffic data using road traffic library of AnyLogic. The experimentation is then conducted to evaluate performance under different signal configurations. The methodology facilitates a low-cost decision-making approach for real-time traffic management. This study expresses the following contributions.

- This study proposes a traffic light model using AnyLogic simulation that adjusts signal timings based on real-time traffic flow.
- A customizable simulation model using DES and road traffic libraries is developed, providing a reliable platform for traffic control system.
- This study presents the traffic performance based on number of vehicles for defined routes for urban traffic planning and smart infrastructure development.

In the upcoming sections, it covers the following aspects of this work. Section II provides an overview of literature review, while Section III introduces the entire process and design of this work. Section IV presents the results and discussion of exploration of this work. Finally, Section V will offer a conclusion summarizing this study.

II. LITERATURE REVIEW

The growing demand for smart transportation systems has driven researchers to explore innovative approaches to improve traffic light control and reduce urban congestion. Multiple studies have utilized simulation-based frameworks, particularly AnyLogic, to appreciate and improve the ability of traffic signal systems.

Pop [13] explored decision-making in traffic intersection coordination. It employed AnyLogic to compare different intersection configurations—traffic lights, roundabouts, and uncontrolled intersections—based on travel time and throughput. This research confirmed that simulation tools could effectively measure how infrastructure changes impact traffic flow, underscoring AnyLogic's adaptability in modelling urban road systems. Liu and Song [14] further demonstrated AnyLogic's capabilities by analysing a congested route to Tianjin Airport. Their agent-based model simulated vehicle movements and signal phase adjustments, revealing a substantial reduction in travel time when signal durations were optimized. The iterative experimentation led to a 9% improvement in average system efficiency. Their work validated the advantage

of integrating real-time traffic dynamics into smart signal algorithms. Sharma and Indu [15] proposed a GPS-enabled adaptive traffic light system that dynamically allocates green time based on data collected via mobile apps and Google APIs. Their method, simulated in AnyLogic, reduced congestion without the need for costly infrastructure like cameras or sensors. It offered a low-cost, scalable model for real-time traffic adaptation but depended heavily on user participation and mobile penetration.

Stupin et al. [16] introduced a hybrid traffic phase optimization model at a T-shaped intersection using macroscopic AnyLogic modelling. Their simulation dynamically adjusted green light durations based on time-of-day vehicle counts. While the approach showed operational improvements, it required manual calibration of traffic parameters, limiting its real-time applicability in larger networks. Benčat and Janota [11] analysed controlled vs. uncontrolled intersections, leveraging AnyLogic's hybrid capabilities. Their study illustrated the usefulness of agent-based and discrete-event modelling in a unified platform. However, it remained focused on a single intersection, lacking multi-intersection coordination analysis, which is crucial in real-world urban settings. Li et al. [17] examined signal phase timing in the context of connected vehicle environments. They found that agent-based simulations could predict reductions of up to 28% in queue lengths and system delays when real-time signal optimization was implemented. However, their reliance on connected vehicle data limits broader applicability in cities without such infrastructure.

Across all reviewed works, several limitations are apparent. Many models are restricted to single intersections or simplified road networks, which do not reflect the complexity of actual traffic systems. Moreover, several studies rely on assumptions or static parameters rather than real-time sensor or GPS data. While AnyLogic proves to be a powerful and flexible tool, integration with real-time data feeds and broader city-wide systems remains underdeveloped. In summary, existing literature confirms that AnyLogic-based traffic simulations can meaningfully inform traffic light optimization. However, gaps remain in real-time adaptability, and multi-intersection coordination. This paper aims to fill these gaps by developing a dynamic, smart traffic light model that can be expanded to larger urban networks while maintaining a low-cost decision-making approach for real-time traffic management.

III. METHODOLOGY

This work adopts a simulation-based approach using AnyLogic to design, model, and evaluate a smart traffic light system at urban intersections. Given the complexity of real-world traffic behaviour—characterized by stochastic vehicle arrivals, varying queue lengths, and changing traffic densities—DES modeling is employed to simulate the individual behaviour of vehicles and their interactions with traffic signal systems. Drawing on insights from prior studies that successfully utilized AnyLogic's Road Traffic Library to replicate realistic traffic flows and optimize signal timings, this research builds a detailed, customizable model that captures micro-level traffic dynamics. Core components include the generation of vehicle agents, traffic light phase controllers, and queuing behaviour at

intersections. Process logic, informed by traffic volume and wait time data, is integrated to adapt signal durations dynamically. The model is validated against number of vehicles for different routes across multiple cases. Simulation results are analysed to determine the system's responsiveness under various traffic conditions. The following sections detail the step-by-step implementation of the smart traffic light system.

A. Road Intersection Model in Traffic Light System

The road intersection model in the proposed smart traffic light system simulates a typical three-road junction with directional vehicle movement from West, East, and South, as depicted in the Fig. 1 model. Vehicles can enter the intersection from any of the three directions and proceed toward other lanes based on route decisions, making the system dynamic. This six-way directional model includes vehicles arriving from West (1), East (3), and South (5), and departing toward East (4), West (2), and South (6), respectively. A pedestrian crossing is also integrated, adding another layer of traffic control logic. The simulation uses AnyLogic's Road Traffic Library to realistically represent vehicle behaviour, signal phase timing, and pedestrian interaction. Each vehicle is modelled as an agent with specific arrival times, speeds, and turning behaviours. The intersection is controlled by the coordinated traffic light system, which alternates between red, green, yellow and pedestrian crossing phases based on queue lengths and signal timing logic. The smart traffic light model for road intersection incorporates an DES where each vehicle and pedestrian interacts with traffic lights and responds to signal logic. The simulation is conducted over the specific duration of 3600 seconds. The traffic signal model uses alternating 30-second red or green phases, each follows by a 3-second yellow phase, creating a highly structured and repetitive traffic signal cycle. These durations are assigned to lane connectors to reflect controlled traffic flow transitions, enabling safe turning and merging behavior in a simulated road intersection. This allows for evaluation of intersection performance under varying traffic volumes, supporting real-time adaptation of green light phases. The model aims to reduce travel times, optimize traffic flow, and ensure pedestrian safety through a coordinated and smart signal control system.



Fig. 1. Road (3-Way) Intersection Model with Traffic Light (2D View).

B. Library Blocks and Process Flowchart for Road Traffic Light Model

The road traffic library enables the simulation and modelling of vehicles navigating a road network. This library provides behaviour closely resembling real-world vehicle movement, incorporating priority rules, speed control, and lane selection strategies to optimize travel times. It facilitates realistic traffic flow by allowing vehicles to prioritize routes and choose lanes with less congestion to minimize travel duration.

The process flowchart in Fig. 2 visually represents a detailed road traffic light model incorporating multi-directional traffic flows and heterogeneous vehicle types such as cars and buses. The system is divided based on entry directions: vehicles (both cars and buses) enter from West, East, and South. Each incoming traffic stream is connected to corresponding paths that guide the vehicles toward one of three destinations—West, East, or South—based on pre-defined route logic. The flow begins with source blocks for each vehicle type and direction. In this case, if “Bus from West” and “Car from West” are generated independently, then proceed through control blocks (likely carMoveTo) based on their assigned routes. Similarly, vehicles from the East and South are processed. These directional flows are then merged and regulated by a central traffic light logic system that coordinates vehicle movement across the intersection.

The model ensures that only one direction moves at a time, preventing conflicts and reducing travel times. The “Vehicle Dispose” block represents the end of the journey for each vehicle, ensuring efficient removal from the simulation. The flowchart emphasizes synchronization, vehicle-type handling, and directional flow control ensuring a realistic and modular traffic simulation. By incorporating traffic light logic with directional movement and disposal points, the model can appreciate the performance of congestion, delay time, and signal potency under various traffic loads.

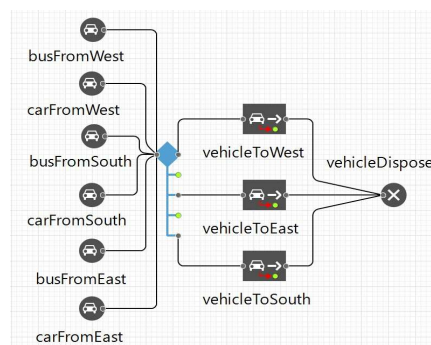


Fig. 2. Process Flowchart of Traffic Light Model for Vehicles.

C. Traffic Flow Decisions for Vehicles in Different Stages

Fig. 3 illustrates the explanation of the coordination of traffic flow decisions for vehicles in different stages. Vehicles enter the intersection from west, east, and south, and depart toward west, east, and south, respectively.

In Fig. 3(a), a bus approaching from the west makes a left turn to proceed westward, while a car from the east also makes a left turn to head eastward. Other incoming vehicles from the west, east, and south, positioned in different lanes, are visible

waiting behind the stop line, highlighting the traffic flow process. In Fig. 3(b), a car approaching from the east makes a left turn to move southward along the green arrow. Fig. 3(c) represents yellow arrow indicators, where vehicles are slowed down and prepared to stop if it is safe to do so. It signals that the light is about to change to red. In Fig. 3(d), a car from the south turns left to proceed westward, while another car from the south crosses the intersection, heading eastward along the path indicated by the green arrow.

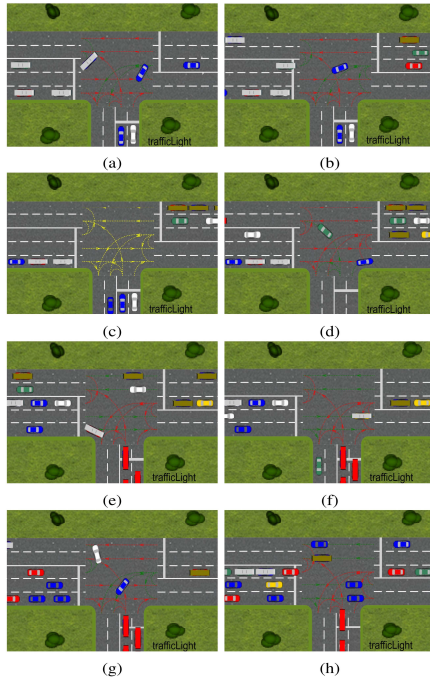


Fig. 3. Coordination of Traffic Flow Decisions for Vehicles in Different Stages (2D View).

In Fig. 3(e), two vehicles make a straight turn from east to west and a bus moves from west to south. In addition, several other vehicles are waiting behind the stop line in different lanes. As shown in Fig. 3(f), through the intersection, vehicles coming from the west will go eastward in their lane as seen in the scene of a bus moving forward and vehicles from the east will proceed westward. In Fig. 3(g), a car approaching the intersection from the east continues southward in its designated lane, following the path indicated by the green arrow, while another car from the west proceeds westward, maintaining its direction through the intersection. Moreover, in Fig. (h), two cars passing from west to eastward while another two cars are moving from east to westward, along the green arrow.

In this model, congestion primarily occurs at the Southbound entry lane, which experiences higher average queue lengths and vehicle delays compared to the East and West directions. This is mainly due to cross-directional turning movements, especially when vehicles from the South attempt to turn left while vehicles from the East and West proceed straight or turn right. In another, the Eastbound and Westbound flows maintain lower queue levels due to more consistent green phases and less complex routing. The smart model improves flow by dynamically

adjusting signals, reducing queues, and increasing the number of vehicles disposed.

D. 3D View of the Road Traffic Light Model

In this work, the visualization of 3D view for a traffic light model illustrates in Fig. 4 as one sample. The 3D view of the traffic light model offers a dynamic and spatially detailed representation of vehicle interactions and traffic signal operations within a simulated urban intersection. In this model, the layout features directional flows from the West, East, and South, where both cars and buses are generated and routed through a multi-lane intersection regulated by intelligent traffic signals. The integration of 3D modelling allows for enhanced visualization of vehicle behaviors such as queuing, turning movements, acceleration, and stopping at traffic signal points.

Traffic lights are positioned logically at entry lanes and transition phases between red, yellow, and green, ensuring vehicles follow coordinated signal phases. Buses, due to their size and longer turning radius, are distinctly represented in the simulation to demonstrate priority movement and designated turning strategies. Vehicles, on the other hand, exhibit more frequent lane changes and responsive movements based on traffic signal timing. The simulation dynamically reflects traffic density, where the 3D objects react to the active signal logic defined in the model. The realism in vehicle spacing, stop-line adherence, and movement animations provides users with an immersive experience, allowing for close observation of congestion buildup, traffic resolution, and light-controlled interactions. This view not only validates the behaviour of individual vehicle agents but also helps evaluate the effectiveness of signal timing strategies and directional traffic coordination at a visual level. The model serves as a vital tool for analysing real-time traffic behaviours and optimizing the intersection design for efficient vehicle flow and minimal delays.



Fig. 4. 3D View of a Traffic Light Model Related to the Vehicles for one Case.

IV. RESULT ANALYSIS

This section presents the simulation results and analysis of the proposed smart traffic light model developed using AnyLogic. The model simulates a three-way intersection with traffic flows from the west, east, and south directions involving both car and bus agents as vehicles. The simulation focused on evaluating key traffic performance as the number of vehicles with movement activity under various traffic conditions. Coordinated signal timing, directional vehicle routing, and adaptive light phase logic has been implemented to replicate real-world dynamics. The visual outputs, combined with statistical data, illustrate how traffic signal adjustments affect

TABLE II. ANALYSIS OF NUMBER OF VEHICLES FOR DEFINED ROADS

Time (Sec)	Vehicle Generation		Total Generated	Vehicle Routing		Queued	Passed	Total Dispose
32.74	VfW	16	35	VtW	13	11	2	5
	VfS	5		VtS	14	14	0	
	VfE	14		VtE	8	5	3	
46.20	VfW	20	49	VtW	16	11	5	11
	VfS	10		VtS	20	17	3	
	VfE	19		VtE	13	10	3	
50.78	VfW	22	55	VtW	17	12	5	12
	VfS	12		VtS	25	21	4	
	VfE	21		VtE	13	10	3	
61.25	VfW	27	65	VtW	22	14	8	17
	VfS	13		VtS	29	23	6	
	VfE	25		VtE	14	11	3	
69.00	VfW	31	70	VtW	24	14	10	21
	VfS	13		VtS	31	25	6	
	VfE	26		VtE	15	10	5	
86.00	VfW	40	81	VtW	25	11	14	29
	VfS	13		VtS	35	26	9	
	VfE	28		VtE	21	15	6	
104.25	VfW	43	84	VtW	26	9	17	37
	VfS	13		VtS	36	22	14	
	VfE	28		VtE	22	16	6	
122.60	VfW	49	90	VtW	28	10	18	42
	VfS	13		VtS	39	24	15	
	VfE	28		VtE	23	14	9	
136.75	VfW	56	97	VtW	30	10	20	48
	VfS	13		VtS	41	23	18	
	VfE	28		VtE	26	16	10	

congestion and flow stability. The 3D visualization further supported spatial validation and movement logic for each agent. This analysis aims to represents the effectiveness of the intelligent coordination strategy for minimizing the vehicle movements delay in a congested urban intersection environment.

The simulation of this work has been conducted using AnyLogic 8.9.3 Personal Learning Edition installed on a Windows 11 system. The testing machine was a LAPTOP-2DHIQ60K equipped with a 13th Gen Intel(R) Core (TM) i7-1355U CPU at 1.70 GHz, 20.0 GB RAM, and a 64-bit operating system with an x64-based processor. The computing environment provided sufficient processing power for real-time animation, multi-agent interaction, and iterative optimization across different traffic density scenarios. The simulation run settings has been configured with warm-up time, consistent time units, and predefined intervals to ensure repeatability and accuracy in result collection.

TABLE II. SUMMARY OF NOTATIONS

Defined Notation	Description
VfW	Vehicles from West
VfS	Vehicles from South
VfE	Vehicles from East
VtW	Vehicles to West
VtS	Vehicles to South
VtE	Vehicles to East

Moreover, Table I provides a concise summary of notations representing vehicle movements for routes from the east, west,

and south, providing a clear description of traffic flow patterns to evaluate the performance of the smart traffic light system at different stages. It categorizes vehicles based on their origin and destination directions. The notations VfW, VfS, and VfE refer to vehicles originating from the west, south, and east, respectively. Similarly, VtW, VtS, and VtE denote vehicles heading toward the west, south, and east. These notations simplify the representation of vehicle trajectories, enabling clear and efficient communication of traffic flow patterns.

In addition, Table II demonstrates an analysis of vehicle dynamics across defined routes at various time intervals, focusing on traffic generation, routing, queuing, passing, and disposal. The table organizes data into key categories: the time (in seconds), the number of vehicles generated per route, total generated vehicles, vehicle routing dynamics, and the corresponding queuing, passing, and disposal figures for each route. The routes are denoted as vehicles from the west (VfW), south (VfS), and east (VfE), along with respective destinations: west (VtW), south (VtS), and east (VtE). The data showcases a time-based progression where the total number of vehicles generated steadily increases, reflecting an escalating traffic load. As seen in the running simulation of road traffic light model, 35 vehicles were generated at 32.74 seconds, while at 136.75 seconds, the count rose to 97. Each route highlights the flow of vehicles, detailing how many queued, passed, and has been disposed of. The model scenario running at 32.74 seconds shows that 16 vehicles are coming from the westward (VfW), 5 from VfS and 14 from VfE, while 13 vehicles are going to westward (VtW), 14 to VtS and 8 to VtE. In this case, 11 vehicles are queuing for VtW, 14 for VtS, and 5 for VtE, where 2 vehicles for VtW, 0 for VtS, and 3 for VtE are passing. The analysis underscores the variations in queuing and passing efficiency for

different routes. In another scenario of running simulation at 136.75 seconds, total 97 vehicles were generated from VfW, VfS, and VfE, and 48 vehicles were disposed to VtW, VtS, and VtE.

It reveals that while the westward route consistently generates the vehicle count, the eastward and southward routes maintain balanced flows. The table is instrumental in evaluating the performance of smart traffic control systems, and presenting strategies for improved traffic flow. Moreover, Fig. 5 depicts the number of vehicles generated and disposed of over time (in seconds) in the simulation. It shows a steady increase in vehicle generation over time, while disposal rates lag initially but gradually align, reflecting the system's processing efficiency as time progresses. Over time, as traffic density increases, queuing and disposal rates fluctuate, indicating the system's ability to adapt to congestion and maintain flow continuity. This structured data provides valuable insights into traffic flow management, emphasizing the performance of traffic routing mechanisms under varying conditions. It highlights the balance between generated traffic and the system's capacity to process and route vehicles efficiently across different directions.

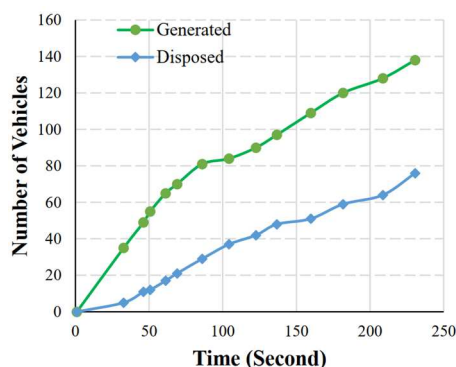


Fig. 5. The graph of Vehicle Generated and Disposed in Traffic Light Model.

V. CONCLUSION

This study successfully developed and simulated a smart traffic light system using the AnyLogic platform to address inefficiencies in conventional fixed-timing traffic signals. The model demonstrated the potential of discrete event simulation to systems and reduced the vehicle's travel time, validating the effectiveness of the coordination approach in traffic signal. Furthermore, the modular design and use of AnyLogic's Road Traffic Library provides vehicle queue and traffic flow patterns by adjusting traffic light signal for various urban scenarios. The 3D model visualization reinforced user understanding and decision-making through immersive observation of traffic behaviours under different conditions. This research contributes an effective simulation model for low-cost, smart traffic management that can be implemented in large cities. Future work can extend the model to include multi-intersection coordination, integration with IoT-based real-time sensors, and machine learning algorithms for predictive traffic control. Overall, this simulation-based approach supports the vision of sustainable, smart urban infrastructure aimed at minimizing traffic delays and environmental impact.

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